

AI-Enhanced Event Management Platform

Detailed project brief generated from the project record.

CANDIDATE

Rahul Sharma

DURATION

90 days

TECHNOLOGY STACK

JavaScript

TypeScript

React

Next.js

Node.js

Express.js

Laravel

Tailwind CSS

Python

HTML

MySQL

Groq API

AWS

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

A full stack web app where organizers create events and sessions, attendees browse schedules, and an AI service suggests personalized session lists. The project evaluates API design, authentication, database modeling, AI integration, PDF export, and deployment.

CANDIDATE

Rahul Sharma

TYPE

fullstack

DURATION

90 days

Objectives

- Implement role based JWT authentication for attendees, organizers, and admins.
- Create CRUD APIs for events, sessions, and registrations.
- Integrate Groq LLM to generate a personalized session list per attendee.
- Export a printable PDF schedule for any event.
- Deploy a production ready version on AWS with CI/CD.

Candidate Details

Name Rahul Sharma

Technology Stack

Languages, frameworks, databases and tools used for implementation.

JavaScript

TypeScript

React

Next.js

Node.js

Express.js

Laravel

Tailwind CSS

Python

HTML

MySQL

Groq API

AWS

Functional Requirements

Requirement 1

Attendees can register, log in, and view events.

Requirement 2

Organizers can create, edit, and delete events and sessions.

Requirement 3

System returns AI generated session recommendations for a logged in attendee.

Requirement 4

Users can download a PDF of their selected sessions.

Requirement 5

Admins (Laravel) can view usage analytics and manage users.

Requirement 6

All APIs validate input and return appropriate HTTP status codes.

Non-Functional Requirements

Requirement 1

Passwords stored with bcrypt hashing.

Requirement 2

JWT tokens must expire after 1 hour; refresh token flow optional.

Requirement 3

API response time < 300 ms for standard CRUD operations.

Requirement 4

Data must be persisted in MySQL with ACID compliance.

Requirement 5

Application must be responsive and pass basic accessibility checks.

System Modules

Major functional areas of the project.

01**Authentication**

JWT based login, role assignment, password hashing.

- Register
- Login
- Role based access middleware

02**User Management**

Profile view/edit, role administration (admin only).

- View profile
- Edit profile
- Assign roles

03

Event & Session CRUD

Organizers manage events and their sessions.

- Create event
- Update event
- Delete event
- Manage sessions

04

AI Recommendation

Call Groq API to generate personalized session list.

- Fetch attendee interests
- Request recommendation
- Store results

05

PDF Export

Generate printable schedule PDFs using server side rendering.

- Select sessions
- Generate PDF
- Download

06

Admin Dashboard (Laravel)

Separate admin UI for analytics and user oversight.

- View event stats
- Manage users
- Export reports

Implementation Roadmap

Recommended sequence for implementing the project.

PHASE 1**Foundation**

Set up repo, CI pipeline, and basic project scaffolding.

Tasks: Initialize monorepo with Yarn workspaces • Configure ESLint/Prettier • Set up GitHub Actions for lint & test • Create Dockerfiles for frontend and backend
Deliverables: Git repo URL • Docker compose file • README with setup instructions

PHASE 2**Authentication & User Management**

Implement secure JWT login and role checks.

Tasks: Design MySQL tables: users, roles • Implement bcrypt password hashing • Create login/register endpoints • Add role based middleware
Deliverables: Auth API • Login UI • Database schema (users, roles)

PHASE 3**Core Event CRUD**

Enable organizers to manage events and sessions.

Tasks: Create MySQL tables: events, sessions, registrations • Build REST endpoints (GET/POST/PUT/DELETE) • Integrate with Next.js pages using SWR • Add client side form validation
Deliverables: Event API • Event management UI

PHASE 4**AI Recommendation Integration**

Provide personalized session suggestions.

Tasks: Write Python micro service that calls Groq API • Expose it via internal HTTP endpoint • Connect Express route to Python service • Display recommendations in frontend
Deliverables: Python service code • Recommendation API • Recommendation UI component

PHASE 5

PDF Export & Admin Dashboard

Add schedule export and admin analytics.

Tasks: Implement server side PDF generation with Puppeteer • Create Laravel admin project • Expose analytics endpoints • Integrate admin UI into Laravel Blade templates
Deliverables: PDF export endpoint • Laravel admin repo • Analytics dashboard

PHASE 6

Testing, Security Hardening & Deployment

Ensure quality and launch on AWS.

Tasks: Write unit & integration tests with Jest & Supertest • Add rate limiting middleware • Configure AWS Elastic Beanstalk for Node and Laravel • Set up environment variables and secrets
Deliverables: Test reports • CI pipeline artifacts • Deployed URL

API Reference

Key endpoints exposed by the project.

METHOD	ENDPOINT	DESCRIPTION
POST	/api/auth/register	Create new user
POST	/api/auth/login	Authenticate and return JWT
GET	/api/events	List public events
POST	/api/events	Create event (organizer)
GET	/api/recommendations	Get AI generated session list
GET	/api/export/pdf	Generate schedule PDF

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.